```
# Tuples are Comparable

print((0,1,2) < (5,1,2)) # 0 is less than 5 - True

print((0,1,200) < (0,3,4)) # 0 is equal to 0 --> 1 is less than 3 - True

print(('Jones', 'Sally') < ('Jones', 'Sam')) # Jones is equal to Jones --> Sally < Sam because l < m - True

print(('Jones', 'Sally') < ('Adams', 'Sam')) # Jones is not equal to Adams but is greater than Adams - therefore, False
```

```
True
True
True
False
```

```
# Concatenation, Repetition, Index

t1=(1,2,3)
t2=(3,'Priya',4)
t3=t1+t2
print(t3)
print()
print(t1*3)
print()
print(t2.index('Priya'))
```

```
(1, 2, 3, 3, 'Priya', 4)

(1, 2, 3, 1, 2, 3, 1, 2, 3)

1
```

```
l=(1,2,3,1,2,3,1,2,3)
print(2 in l)
```

```
True
```

```
# Iterating through a tuple

for name in ('John', 'Kate'):
    print("Hello", name)
```

```
Hello John
Hello Kate
```

```
fruits=('apple', 'banana', 'mango', 'grapes', 'pineapple')
for x in fruits:
    print(x)
```

```
apple
banana
mango
grapes
pineapple
```

Tuples do not support these methods: append remove insert reverse sort

Set

```
fruits={'apple', 'banana', 'mango', 'grapes', 'pineapple'}
print(fruits)

a={1,3,5,2,4}
print(a)
print(type(a))

# RANDOM ORDER

{'grapes', 'mango', 'banana', 'pineapple', 'apple'}
{1, 2, 3, 4, 5}
<class 'set'>

thisSet=set(('apple', 'banana', 'mango', 'grapes', 'pineapple'))
print(thisSet)
print(type(thisSet))

{'grapes', 'mango', 'banana', 'pineapple', 'apple'}
<class 'set'>

a={1,2,2,3,3,3,4,4,4,4}
print(a)

{1, 2, 3, 4}

a={1,2,3}
print(a[1]) # 'set' object does not support indexing

---------------------------------------------------------------------
-----
TypeError                                 Traceback (most recent call
last)
<ipython-input-17-61a87b252975> in <module>
      1 a={1,2,3}
----> 2 print(a[1]) # 'set' object does not support indexing

TypeError: 'set' object does not support indexing

fruits={'apple', 'banana', 'mango', 'grapes', 'pineapple'}
for x in fruits:
```

```
    print(x)
print()
print(len(fruits))

# Once a set is created, you cannot change its item but you can add
new items

grapes
mango
banana
pineapple
apple

5

fruits={'apple', 'banana', 'mango', 'grapes', 'pineapple'}
fruits.add('orange')
print(fruits)

fruits.update(['cherry', 'blackberry', 'dragonfruit'])
print(fruits)

fruits.discard('strawberry') # if the item does not exist, discard()
will not raise an error
print(fruits)

fruits.remove('strawberry') # if the item does not exist, remove()
will raise an error
print(fruits)

{'orange', 'grapes', 'mango', 'banana', 'pineapple', 'apple'}
{'orange', 'cherry', 'grapes', 'mango', 'banana', 'dragonfruit',
'blackberry', 'pineapple', 'apple'}
{'orange', 'cherry', 'grapes', 'mango', 'banana', 'dragonfruit',
'blackberry', 'pineapple', 'apple'}

---------------------------------------------------------------------
-----
KeyError                                  Traceback (most recent call
last)
<ipython-input-19-93efbfaa3e7d> in <module>
      9 print(fruits)
     10
---> 11 fruits.remove('strawberry') # if the item does not exist,
remove() will raise an error
     12 print(fruits)

KeyError: 'strawberry'

fruits={"apple", "banana", "mango", "grapes", "pineapple", "papaya",
"guava", "lemon"}
```

```
print()
fruits.clear()
print(fruits)

del fruits
print(fruits)
```

```
set()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-20-2b95bb775634> in <module>
      6
      7 del fruits
----> 8 print(fruits)

NameError: name 'fruits' is not defined
```

```
fruits={"apple", "banana", "mango", "grapes", "pineapple", "papaya",
"guava", "lemon"}

x=fruits.pop()
print(x)
print(fruits)
```

```
grapes
{'lemon', 'guava', 'mango', 'papaya', 'banana', 'pineapple', 'apple'}
```

```
set1={1,2,3,'b'}
set2={'a', 'b', 'c', 1, 3}
set3=set1.union(set2)
print(set3)

print()

set1.update(set2)
print(set1)

# both union and update will exclude any duplicate values
```

```
{1, 2, 3, 'b', 'c', 'a'}

{1, 2, 3, 'b', 'c', 'a'}
```

```
null_set=set()
print(null_set)

a={1, 2.5, ("pineapple", "pumpkin"), "Jack"}
```

```python
print(a)

b=set("abcd")
print(b) # {'b', 'a', 'd', 'c'}

c={"abcd"}
print(c) # {'abcd'}

print(len(a))

print(("pineapple", "pumpkin") in a) # True
print(("pineapple") in a) # False

print()

for x in a:
    print(x)
```

```
set()
{1, 2.5, ('pineapple', 'pumpkin'), 'Jack'}
{'d', 'b', 'c', 'a'}
{'abcd'}
4
True
False

1
2.5
('pineapple', 'pumpkin')
Jack
```

```python
a_set=set("abcd")
b_set=set("cdef")
print(a_set & b_set) # OR print(a_set.intersection(b_set))

print()

print(a_set - b_set) # OR print(a_set.difference(b_set)) # {'b', 'a'}
print(b_set - a_set) # OR print(b_set.difference(a_set)) # {'e', 'f'}

print()
print(a_set ^ b_set) #{'f', 'e', 'a', 'b'}
# OR b_set.symmetric_difference(a_set) # {'a', 'b', 'e', 'f'}
```

```
{'d', 'c'}

{'b', 'a'}
{'e', 'f'}

{'f', 'b', 'e', 'a'}
```

```
my_set={'a', 'b', 'c'}
print(my_set)

print()

my_copy=my_set.copy()
print(my_copy)

print()

my_ref_copy=my_set
print(my_ref_copy)

print()

my_set.remove('b')
print(my_set) # {'a', 'c'}
print(my_copy) # {'b', 'a', 'c'}
print(my_ref_copy) # {'a', 'c'}
```

```
{'b', 'c', 'a'}

{'b', 'c', 'a'}

{'b', 'c', 'a'}

{'c', 'a'}
{'b', 'c', 'a'}
{'c', 'a'}
```

```
# frozenset() Function - making set immutable
a=(1,2,3,4,5,6,7,8,9,10)
#converting tuple to frozenset
fnum=frozenset(a)
print("Frozenset Object is: ", fnum)
```

```
Frozenset Object is:  frozenset({1, 2, 3, 4, 5, 6, 7, 8, 9, 10})
```

Dictionary

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
    "Year": 2025
}

print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2025}
```

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
    "Year": 2025,
    "Year": 2026
}

print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2026}
```

```
x=thisDict["Model"]
print(x)

y=thisDict.get("Model")
print(y)

print(len(thisDict))

z=thisDict["Year"]
print(z)
```

```
BE 6e Batman Edition
BE 6e Batman Edition
3
2026
```

```
x={
    'a':'b',
    'b':2,
    '2':6
}

print(x['2']) # 6
print(x[2]) # KeyError: 2
```

```
6

---------------------------------------------------------------------------
-----
KeyError                                  Traceback (most recent call
last)
<ipython-input-30-da7ca0d41bc8> in <module>
      6
      7 print(x['2']) # 6
----> 8 print(x[2]) # KeyError: 2

KeyError: 2
```

```
thisDict={
    "Brand": "Mahindra",
```

```
    "Model": "BE 6e Batman Edition",
    "Year": 2025
}

thisDict["Year"] = 2026
print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2026}
```

```
for x in thisDict:
    print(x)
```

```
Brand
Model
Year
```

```
for x in thisDict:
    print(thisDict[x])
```

```
Mahindra
BE 6e Batman Edition
2026
```

```
for x in thisDict.values():
    print(x)
```

```
Mahindra
BE 6e Batman Edition
2026
```

```
for x in thisDict.keys():
    print(x)
```

```
Brand
Model
Year
```

```
for x, y in thisDict.items():
    print(x, ":", y)
```

```
Brand : Mahindra
Model : BE 6e Batman Edition
Year : 2026
```

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
    "Year": 2026
}

print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2026}
```

```
thisDict["Color"]="Matt Black"
print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2026,
'Color': 'Matt Black'}
```

```
thisDict.popitem()
print(thisDict)
```

```
{'Brand': 'Mahindra', 'Model': 'BE 6e Batman Edition', 'Year': 2026}
```

```
thisDict.pop("Brand")
print(thisDict)
```

```
{'Model': 'BE 6e Batman Edition', 'Year': 2026}
```

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
    "Year": 2026
}

del thisDict["Model"]
print(thisDict)
```

```
{'Brand': 'Mahindra', 'Year': 2026}
```

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
    "Year": 2026
}

del thisDict
print(thisDict)
```

```
---------------------------------------------------------------------
-----
NameError                                 Traceback (most recent call
last)
<ipython-input-44-af22f29e0d2e> in <module>
      1 del thisDict
----> 2 print(thisDict)

NameError: name 'thisDict' is not defined
```

```
thisDict={
    "Brand": "Mahindra",
    "Model": "BE 6e Batman Edition",
```

```
    "Year": 2026
}

thisDict.clear()
print(thisDict)
```

```
{}
```